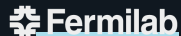

DUNE detector-wide interlocks

What can run together? What must wait? What
needs protection?

Manuel Arroyave

Collaboration review v0.2 – 8 September 2026



Three situations need three different responses

PROTECT EQUIPMENT

LAr level is not confirmed above the minimum

HV is unavailable: voltage-setting, enabling and ramping are blocked.

TAKE TURNS

Camera lights or an external laser with PDS

Block the second request. Confirm the light is off before PDS returns.

KEEP RUNNING; LABEL DATA

Purity monitor with PDS

Keep the photon detection system (PDS) running. Mark the affected data.

Each matrix row should read like a sentence

HOW TO READ IT

**When this happens,
this is what we do.**

Then say why, who is involved, and what lets us return.

Situation	What happens	Before returning
Camera lights are already on; PDS is requested	The new PDS request waits	Confirm light-off and the agreed PDS return checks

Example: O-001. Proposed detector behavior; the exact PDS state still needs agreement.

HV stays unavailable until LAr is above the minimum

HV UNAVAILABLE

LAr level is too low or is not reliably known

Voltage-setting, enabling and ramping commands remain blocked.

LEVEL REQUIREMENT MET

Measured LAr level is above the required minimum

All required instrumentation is covered, including the cathode. Other HV checks must also pass.

TAKEAWAY This is a prerequisite for HV availability. Reaching the level does not switch HV on.

P-001 · Cryogenics + HV + Detector Protection specify the minimum level, reference point and measurement checks.

A fault stops the equipment named in its rule

Equipment problem	Proposed response	Teams to agree it
Abnormal cathode current	Shut down affected HV	HV + Detector Protection
PDS supply fault or overheating	Switch off affected supply or equipment	PDS + power + protection
APA/CRP supply fault	Switch off the agreed output or group	TPC + HV + protection

TAKEAWAY Each team names the equipment at risk and the action needed to protect it.

P-004, P-011, P-012, P-018 · Limits, shared outputs and timing still need agreement.

Camera lights and PDS take turns

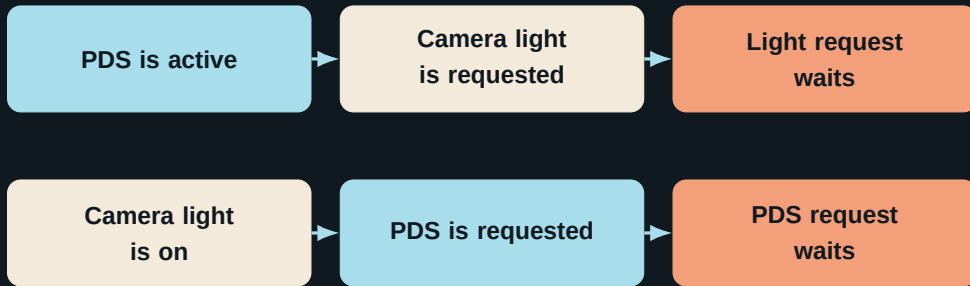


Diagram. The rule applies whichever activity starts first. Confirm illumination is actually off before PDS resumes.

O-001 · PDS must define the exact excluded hardware state and affected region. Old interface wording needs review.

External lasers need an explicit handback to PDS

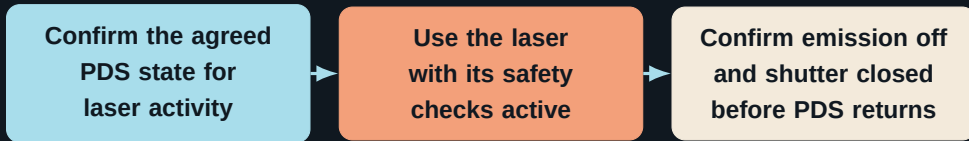


Diagram. PDS and external-laser activity must take turns. The laser team and PDS team agree the return checks.

TEAMS INVOLVED

Laser + PDS + run coordination

Laser personnel protection remains active regardless of the PDS or run state.

O-002, S-002 · NP04 handoffs support this rule; state definitions and timing remain open.

Purity measurements can run while PDS takes data

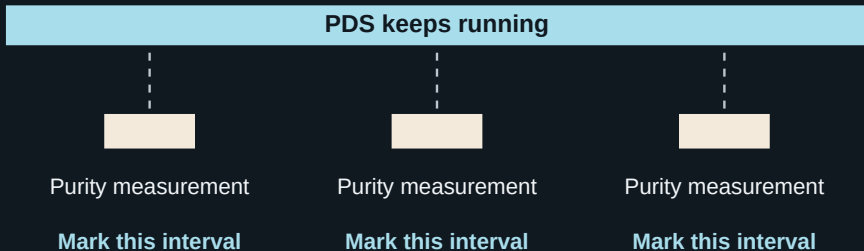


Diagram. The purity-monitor team supplies the times and affected region. Data acquisition records them for analysis.

TAKEAWAY PDS bias stays on for this interaction. Quiet periods can be arranged when a measurement needs them.

Q-001 · This proposal conflicts with the old PDS interface's bias-off text; that text needs correction.

Different light sources have different rules

Activity with PDS	Proposed rule
Camera illumination	Take turns
External ionisation / alignment laser	Take turns; keep laser safety active
Purity monitor	Run together; mark affected data
PDS-owned calibration light	Use the declared calibration procedure
Power-over-fiber for applicable PDS	Required together; keep its own protection active

O-001, O-002, Q-001, Q-002, C-001 · Name the actual source and affected optical region.

TPC teams must agree the allowed detector states

Situation	Question for the detector teams
Cathode HV trips	What must APA/CRP, electronics and affected PDS do?
Mains power disappears	Is a shutdown order needed? Can it still happen?
Filling, draining or a warm test	Which electronics states are allowed in that procedure?

TAKEAWAY These answers remain open. Review horizontal-drift and vertical-drift cases separately.

P-014, P-015, P-019 · Use the approved detector procedure while these questions are resolved.

Every rack rule needs a list of affected equipment

RACK POWER + COOLING

Smoke, water or overheating can stop several systems.

The rack connection list tells us who is affected.

AGREE TOGETHER

Which loads switch off?

Name the rack, supplies and detector equipment.

BEFORE RESTART

What needs inspection?

Confirm cooling, power and the affected equipment before returning.

P-007–P-010 · Rack owners + equipment teams + Detector Protection agree the response and any essential loads kept on.

DAQ needs to know what a detector condition means

Detector condition	What DAQ needs to do
Timing is not ready	Hold steps that require synchronized timing
Equipment state is unknown	Hold unverified changes; show what is missing
An allowed activity affects data quality	Record when, where and which activity

TAKEAWAY Agree in advance whether an ongoing run stops, pauses or continues with a data-quality record.

O-004, O-005, Q-001–Q-005 · The selected run type matters.

An expert test needs a clear handback

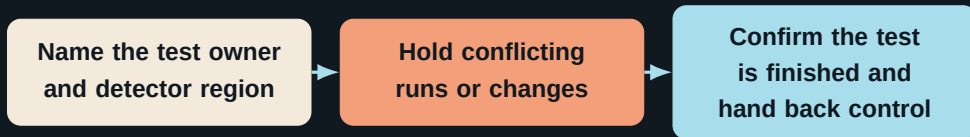


Diagram. ProtoDUNE shifters already coordinated work this way. The detector needs a visible record of who is using each region.

TAKEAWAY The next operator should be able to see who to contact and what still needs checking.

H-001, O-006 · Based on the existing NP02/NP04 evidence review; the recording method remains to be agreed.

Each detector stage needs a clear procedure

INSTALLATION

Who is working?

Record the work area, isolation and the limited tests that are authorized.

COMMISSIONING + INTEGRATION

Who owns the test?

Agree the detector states, affected systems, test duration and return procedure.

RUN

Is this run allowed?

Check equipment readiness and protection. Record conditions that affect the data.

TAKEAWAY Commissioning does not create a general exception to protection or light exclusions.

Missing information needs a specific response

A PROTECTION INPUT IS MISSING

Can we still protect the equipment?

The equipment and protection teams must define whether to block startup or stop the affected equipment.

A PURITY-ACTIVITY RECORD IS MISSING

Can we still identify affected data?

Mark the uncertain data quality. The missing record alone does not require a PDS-bias trip in this proposal.

P-010, P-017, Q-001 · A lost display or connection must be distinguished from a failed protection function.

Clearing an alarm is only the first restart check

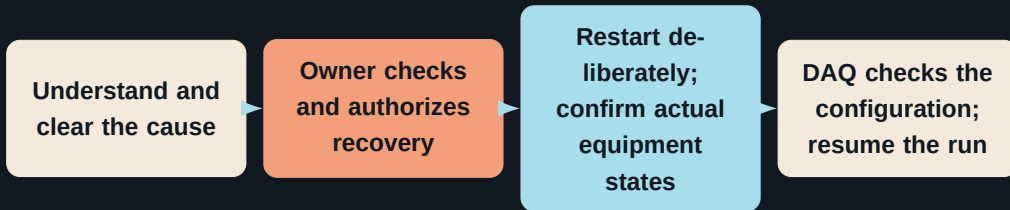


Diagram. Every subsystem supplies its own recovery checks. A returning power supply or network connection does not automatically restart the detector.

O-010 · Agree the people, order and checks for the affected equipment.

Some rules still need a detector decision

Decision needed	Teams to resolve it
Exact PDS state excluded by external light; conflicting old interface text	PDS + light-source teams + DAQ
Response of other systems to cathode or mains-power loss	HV + TPC + PDS + electrical + protection
Electronics during filling; fan, heater and condensation faults	TPC + cryogenics + cooling + protection
Allowed motion and grounding-fault response	Device owners + electrical + detector teams

The companion guide records these questions and the source disagreements. An open row grants no new permission.

Review the rows for your subsystem

WHAT WE NEED FROM EACH TEAM

Tell us what must happen and who needs to agree it.

Start with the detector situation. The integration team can translate the agreed answer into controls and tests.

[Companion guide and review form](#) · Discussion draft throughout.

COMPANION GUIDE

40 rules. 11 team views.

Find your subsystem and review its actions and open questions.

ONE SHORT REVIEW FORM

What happens next?

Name the consequence, response, affected equipment and return checks.